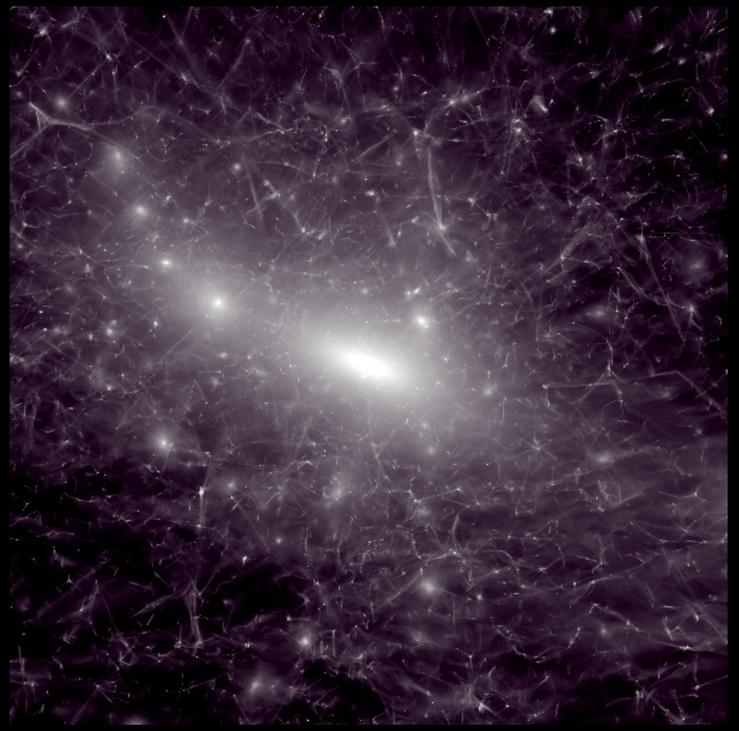
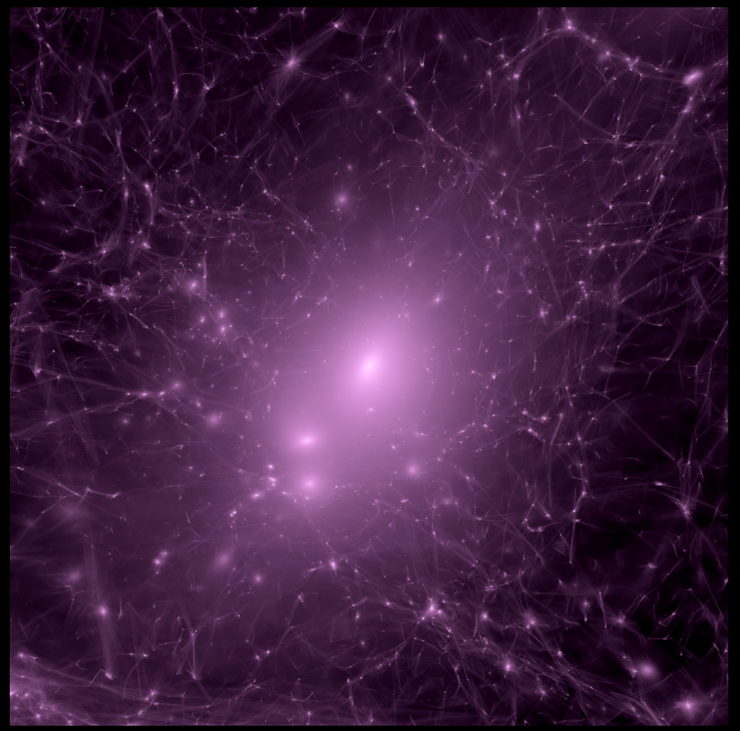
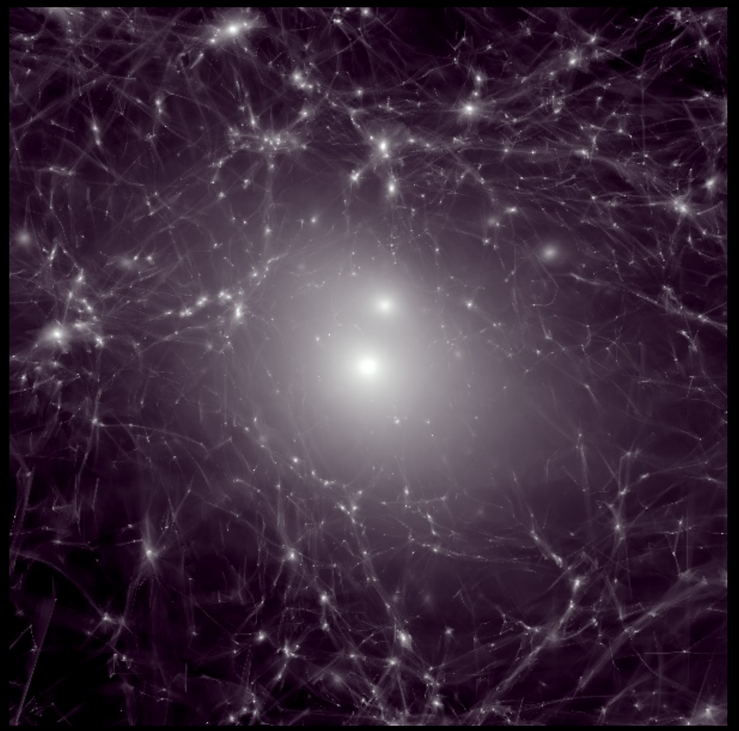
AstroReach SD: Dark Matter & Python







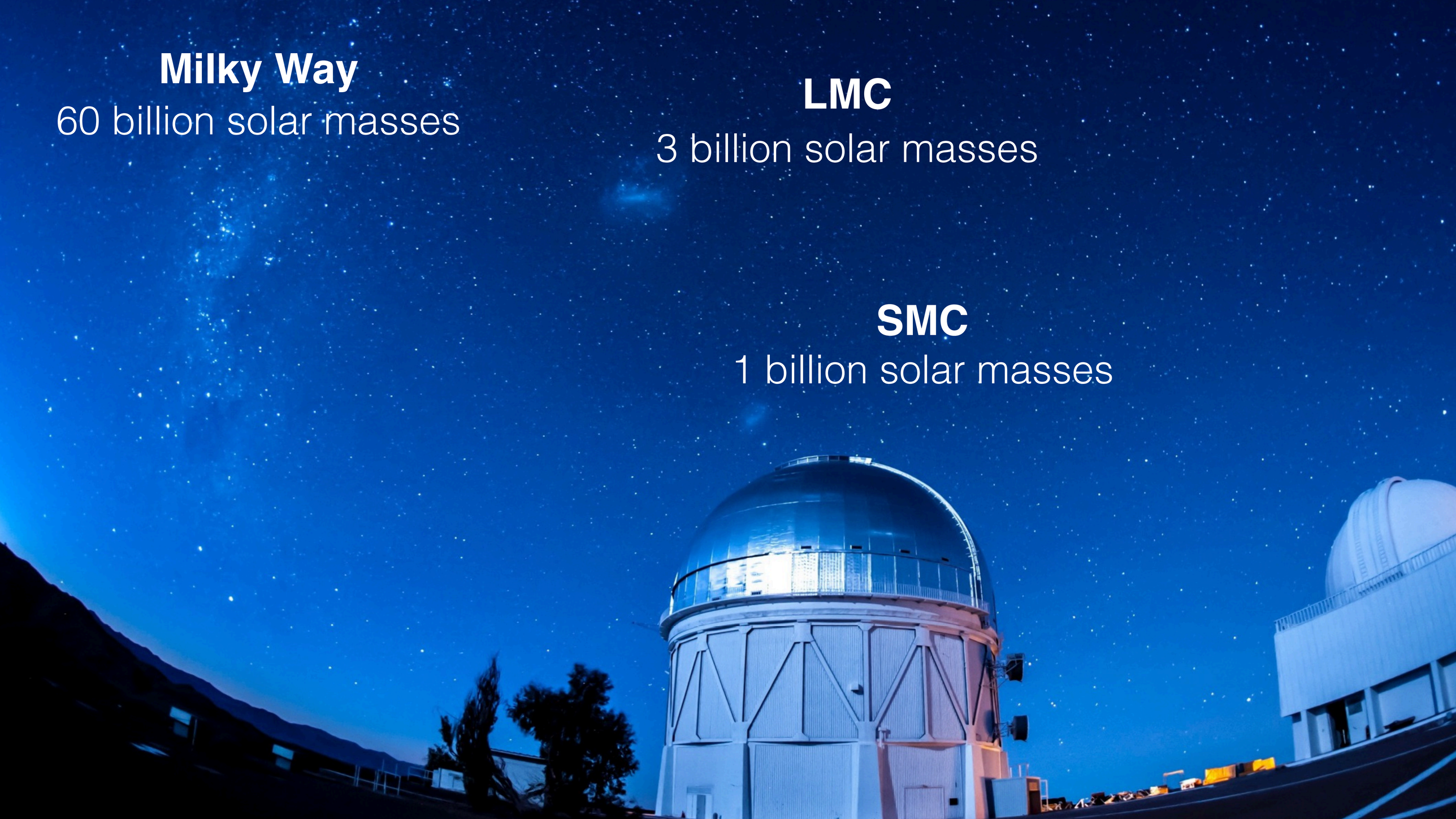
THE PREUSS SCHOOL

Week 5: Dwarf Galaxies & Dark Matter
02/11/2026

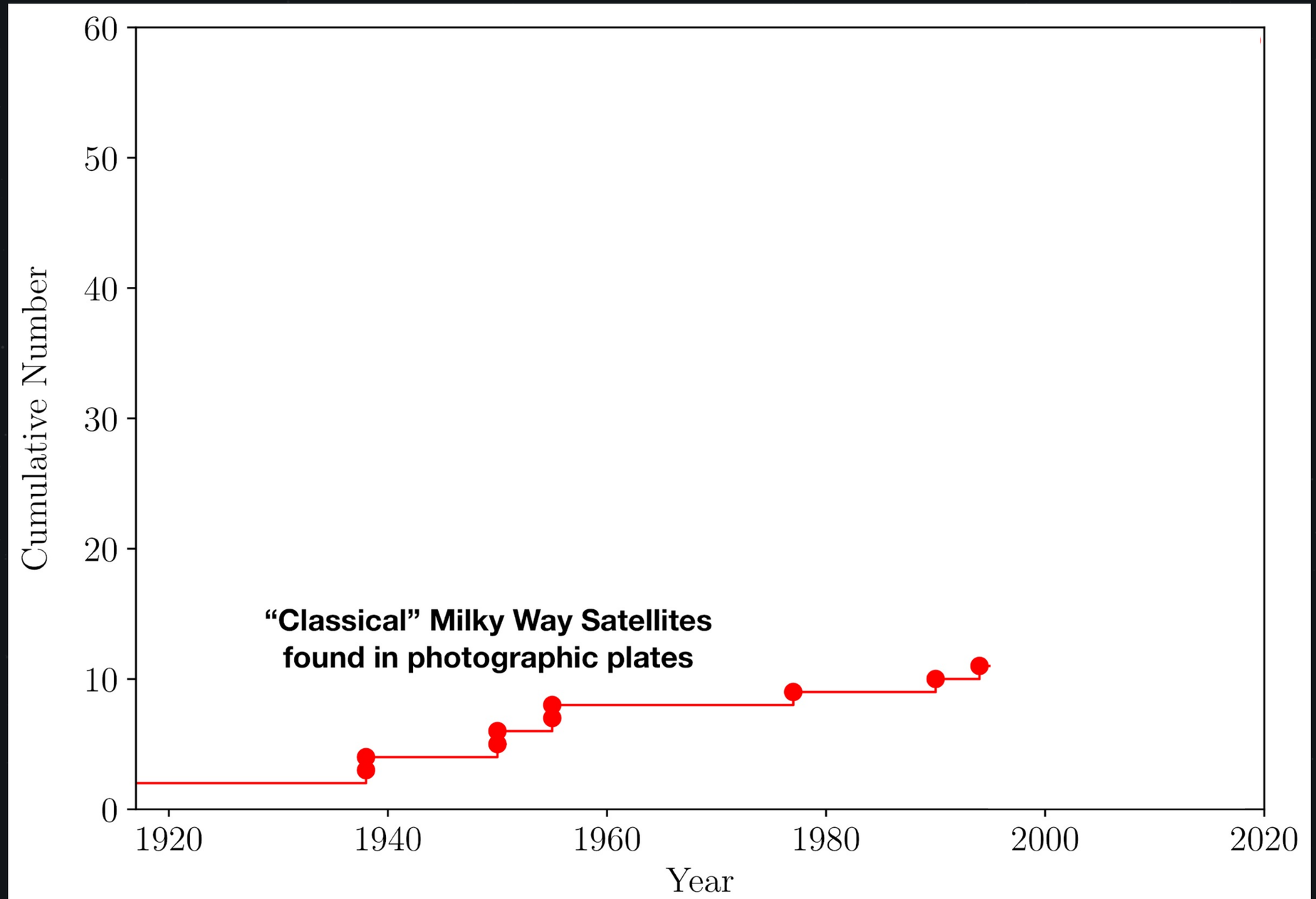
UC San Diego




Milky Way
60 billion solar masses


Milky Way
60 billion solar masses
LMC
3 billion solar masses
SMC
1 billion solar masses

Satellite Galaxies Orbiting the Milky Way



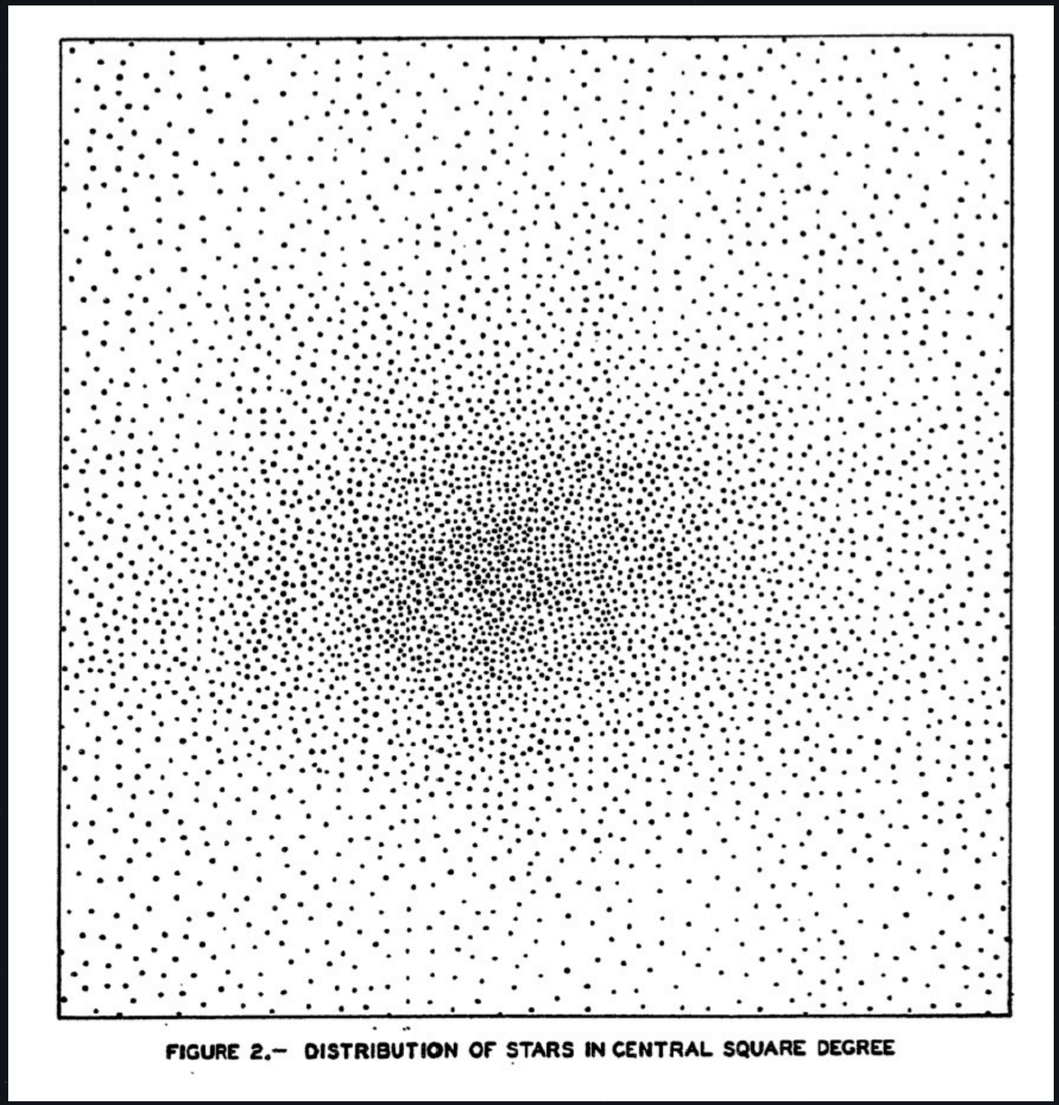


FIGURE 2.— DISTRIBUTION OF STARS IN CENTRAL SQUARE DEGREE

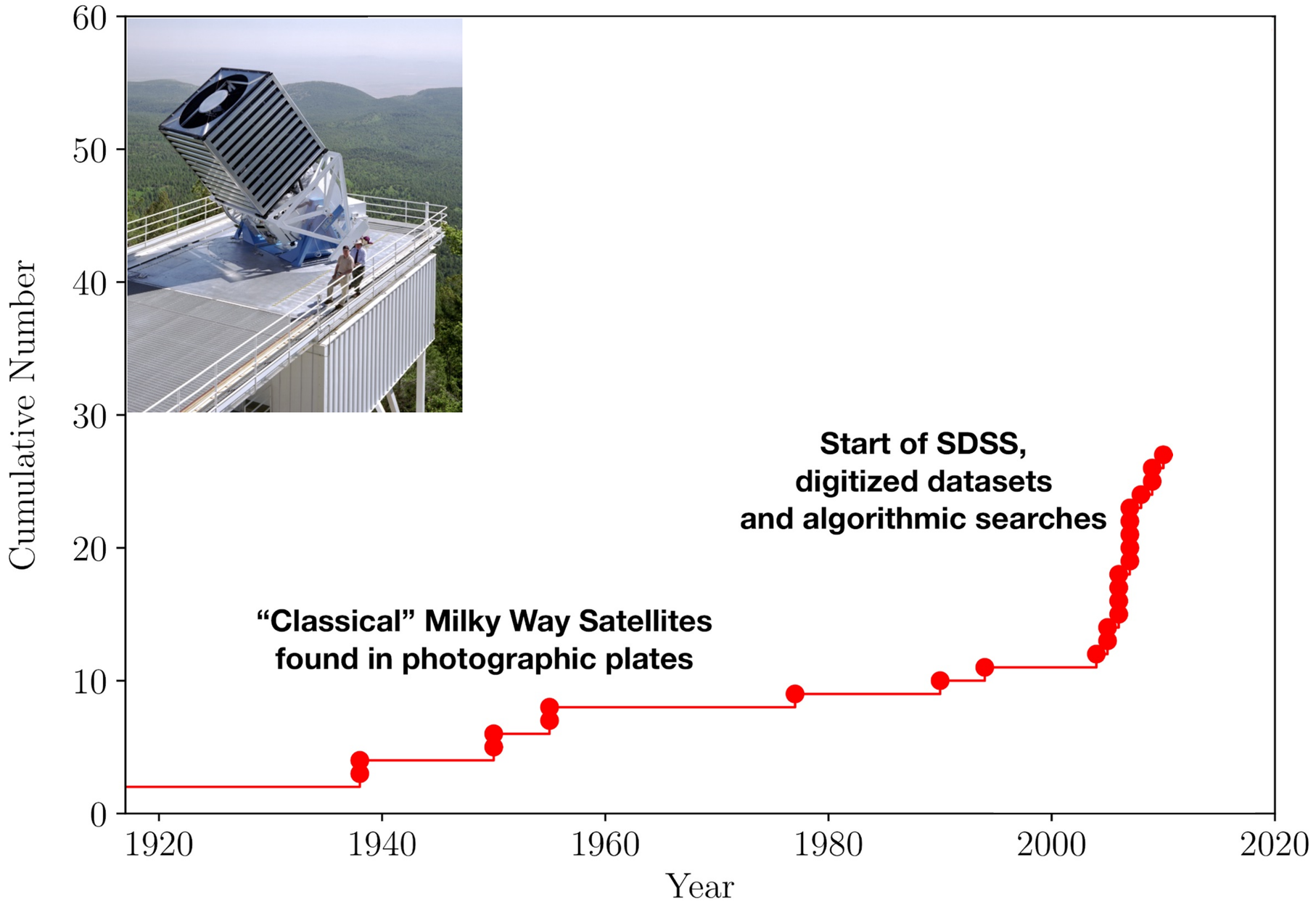
Sculptor

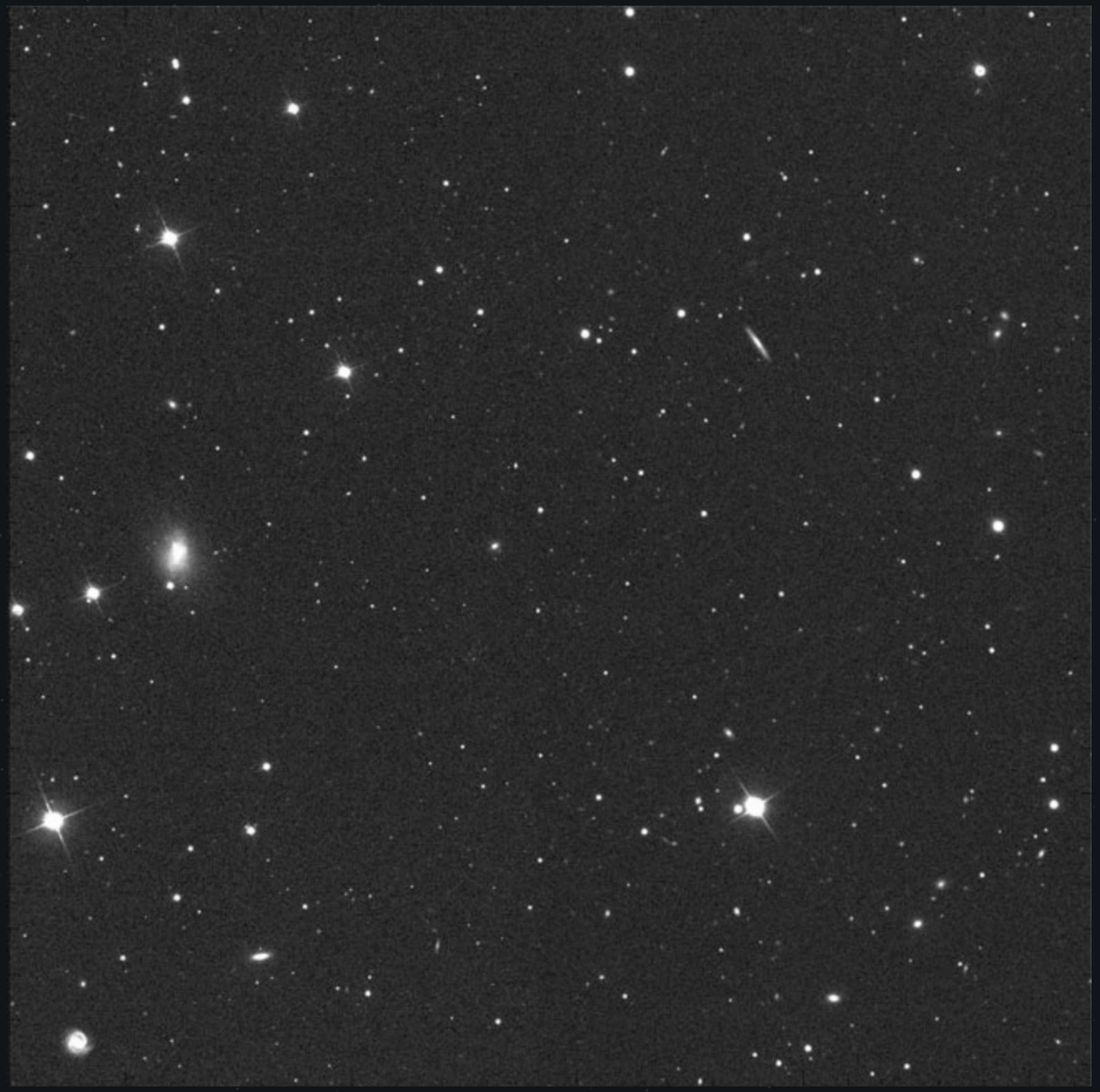
4 million suns

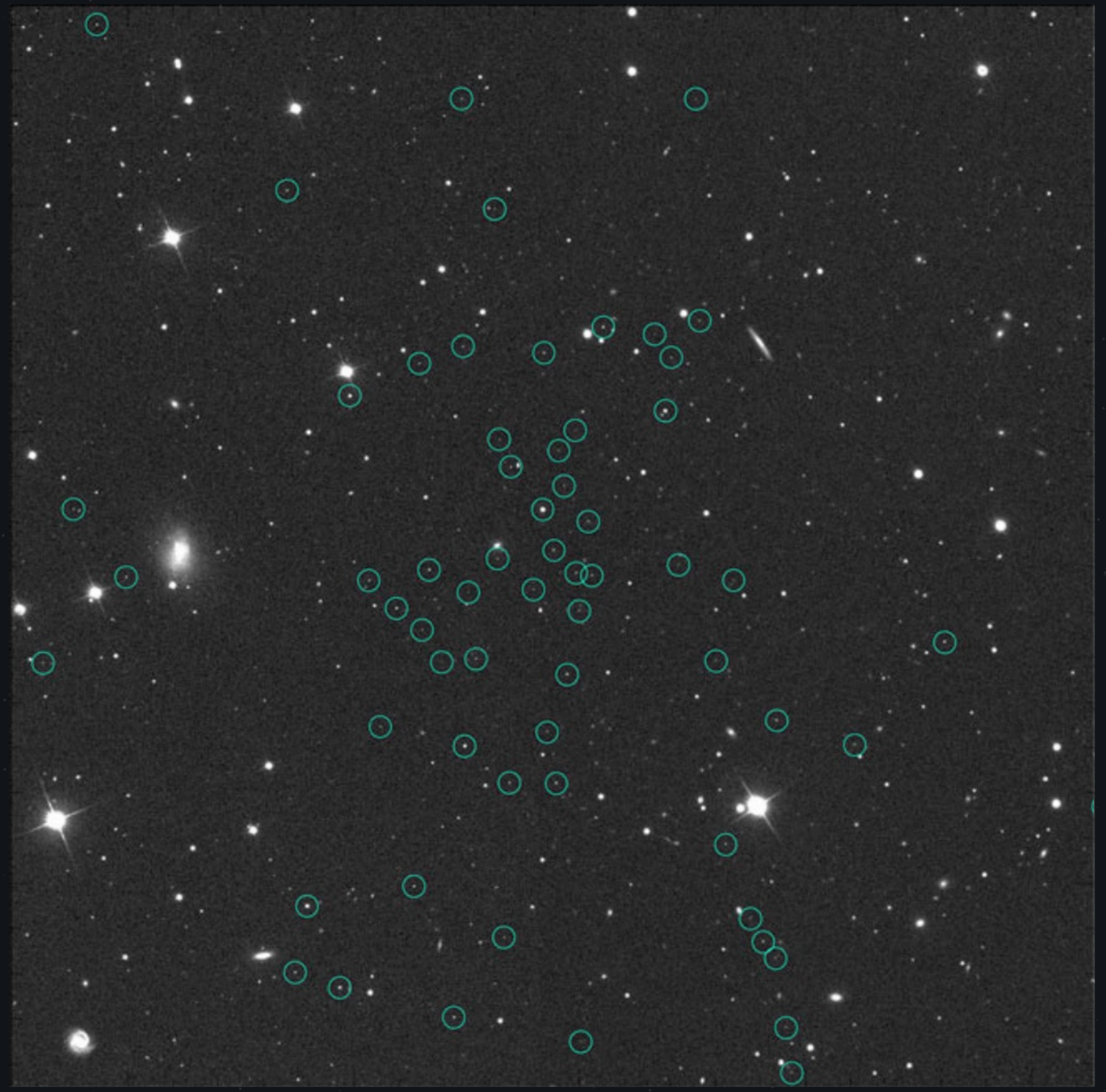
In many respects it appeared to be unlike any known stellar organization. The finding more recently of a similar system in Fornax ... suggests that a description of these objects may be of general interest.

- Harlow Shapley, 1938

Satellite Galaxies Orbiting the Milky Way







Segue I

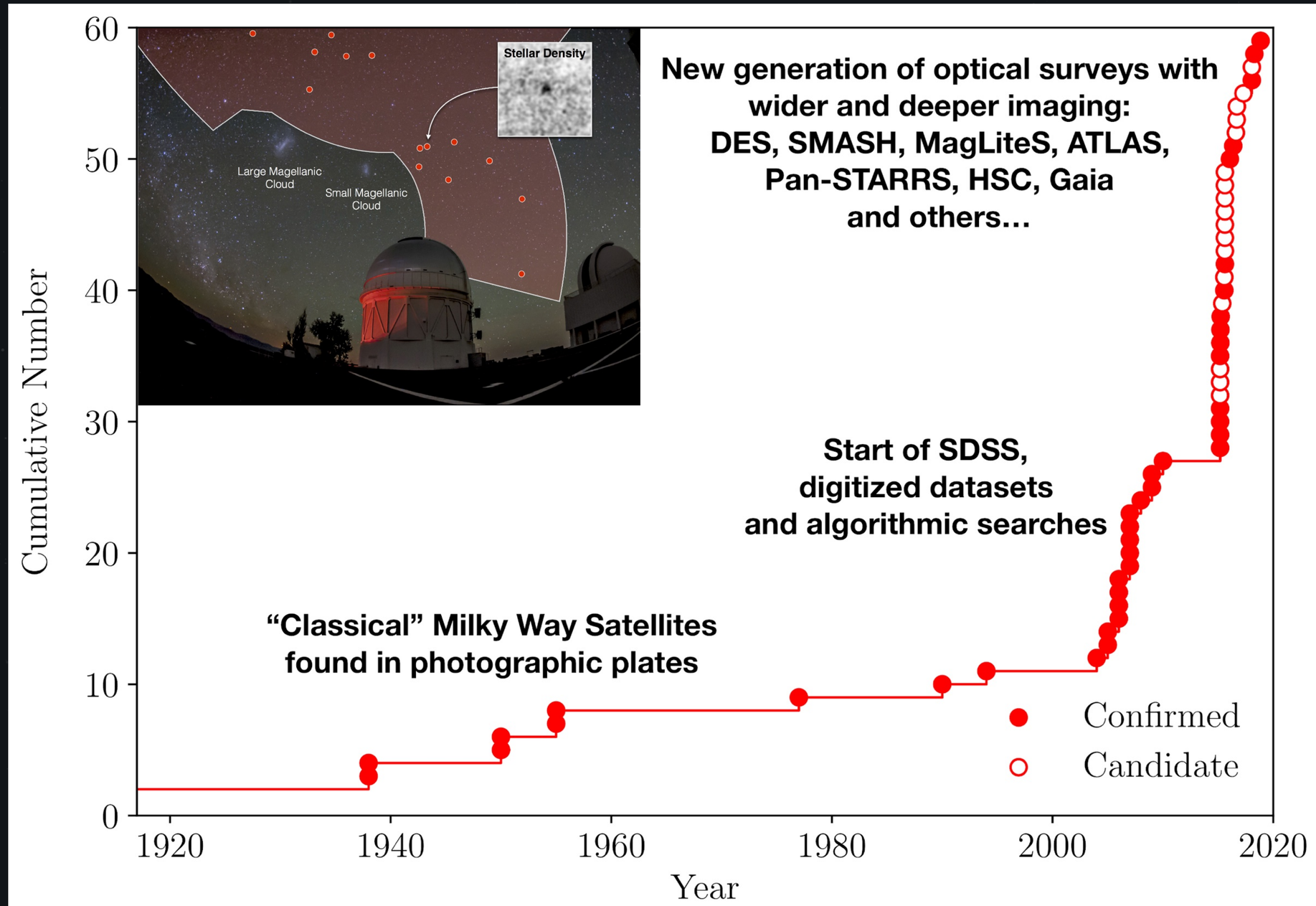
300 suns

the darkest known galaxy!

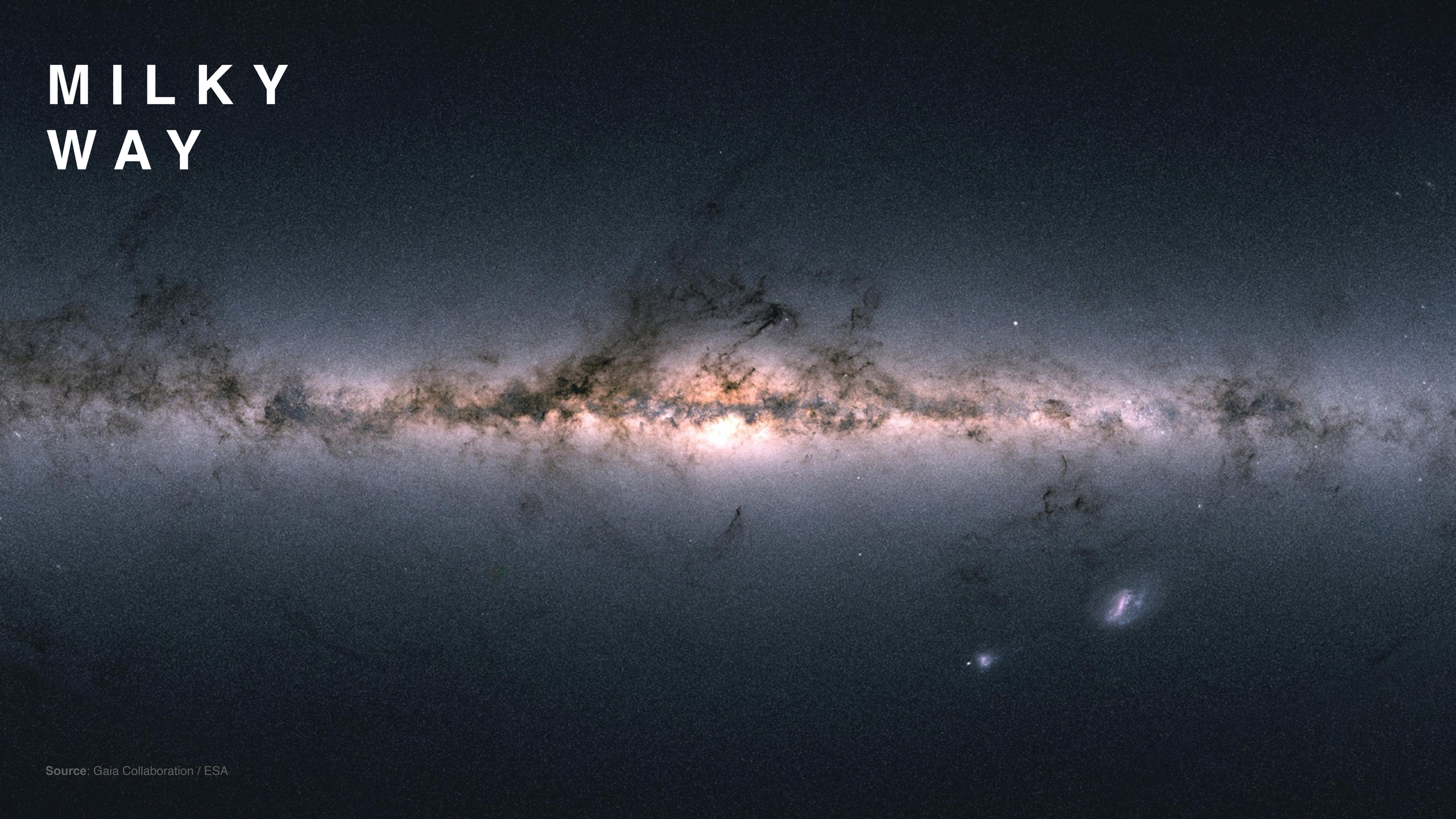
THE DARK ENERGY SURVEY
2012-2018



Satellite Galaxies Orbiting the Milky Way

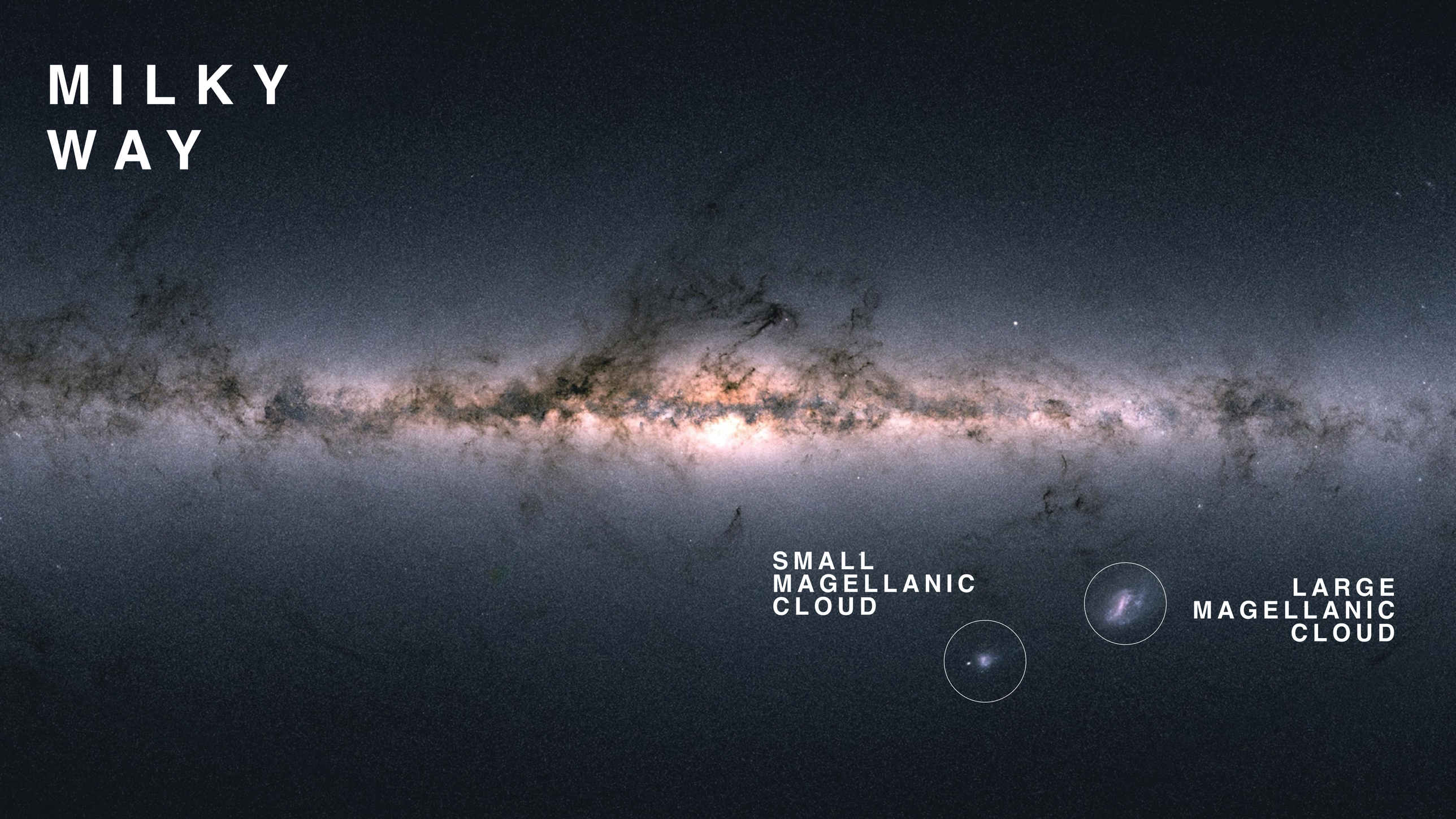


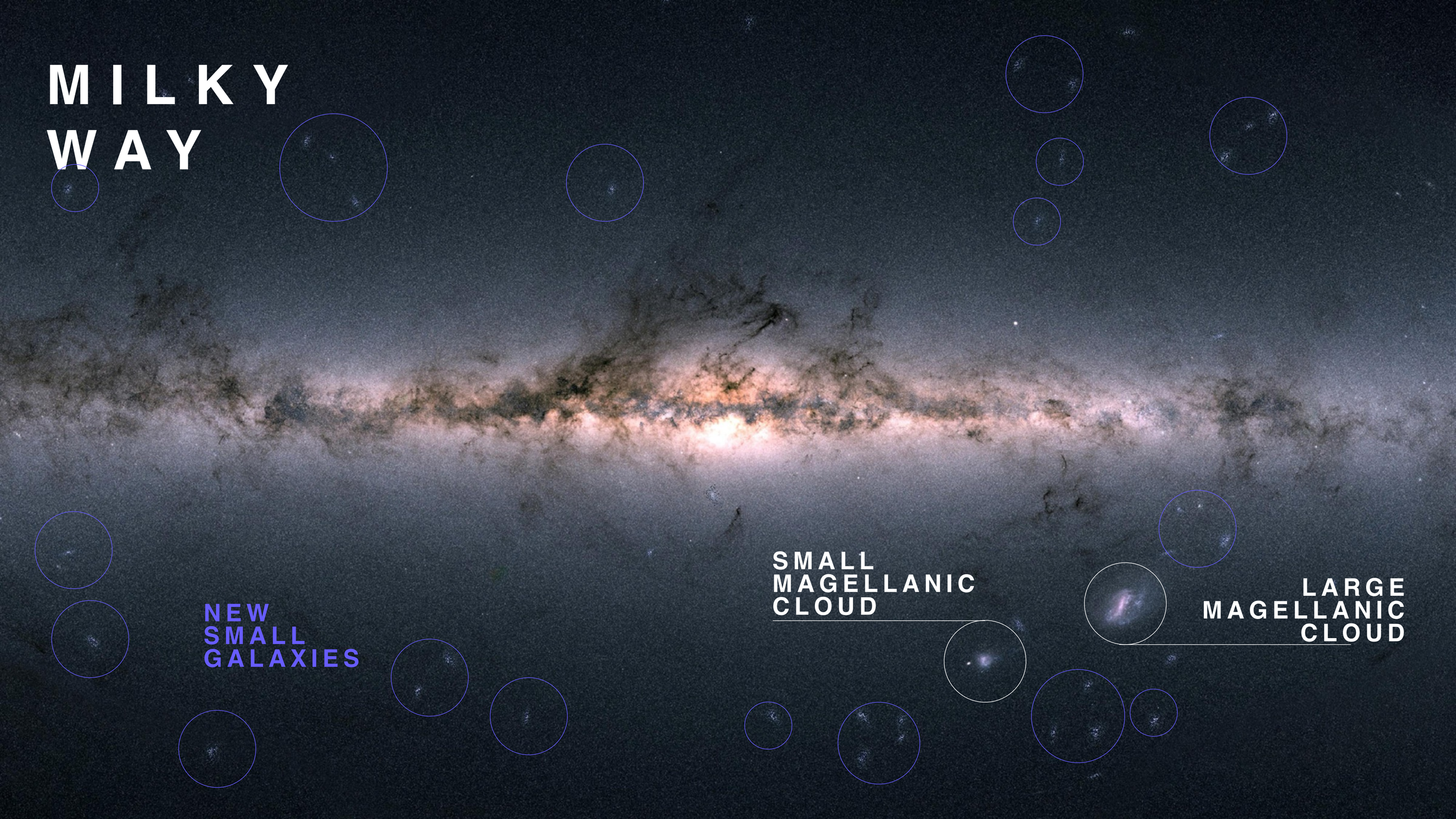
MILKY WAY



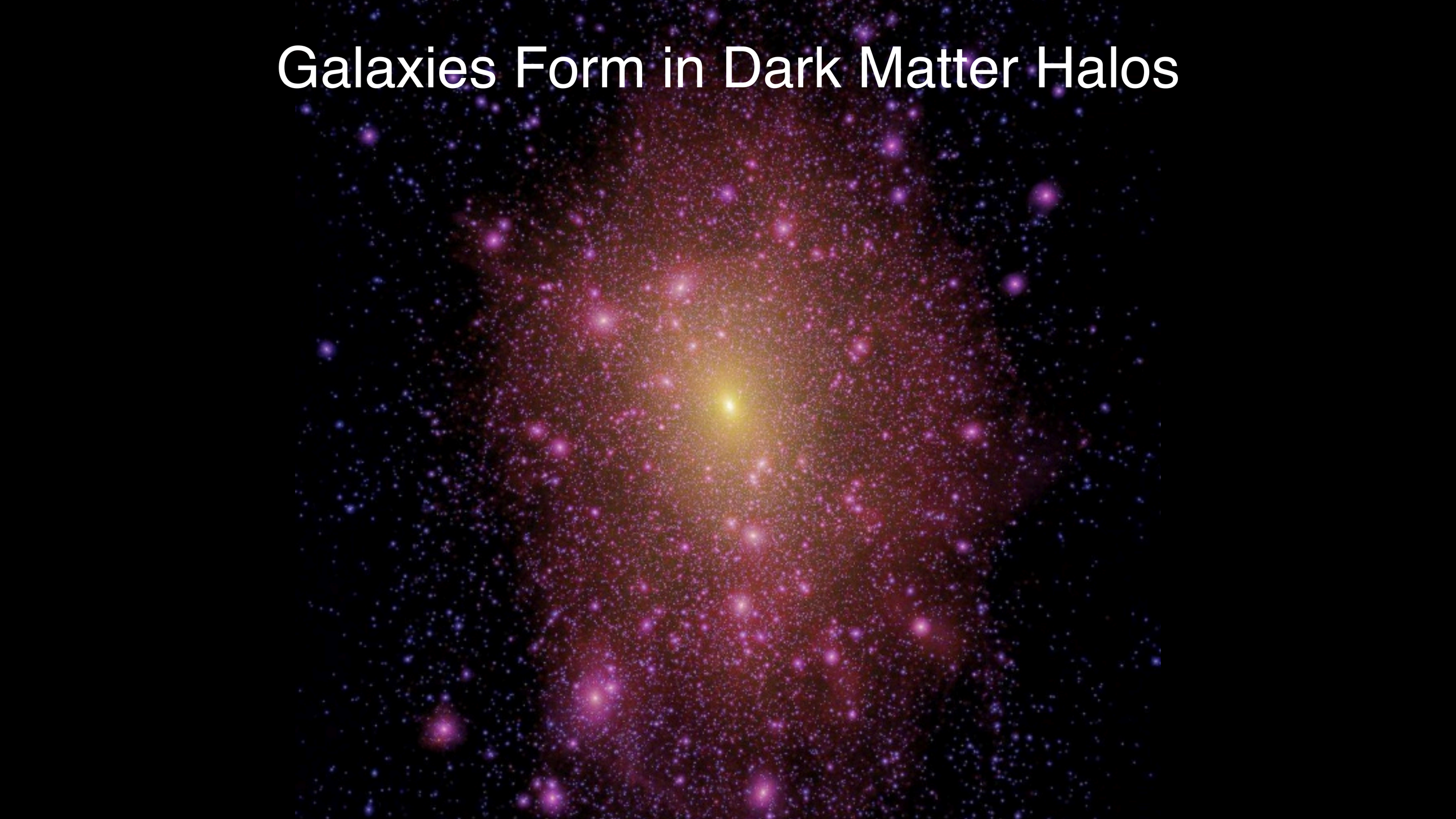
Source: Gaia Collaboration / ESA

MILKY WAY




MILKY
WAY
NEW
SMALL
GALAXIES
SMALL
MAGELLANIC
CLOUD
LARGE
MAGELLANIC
CLOUD

Galaxies Form in Dark Matter Halos



Galaxies Form in Dark Matter Halos



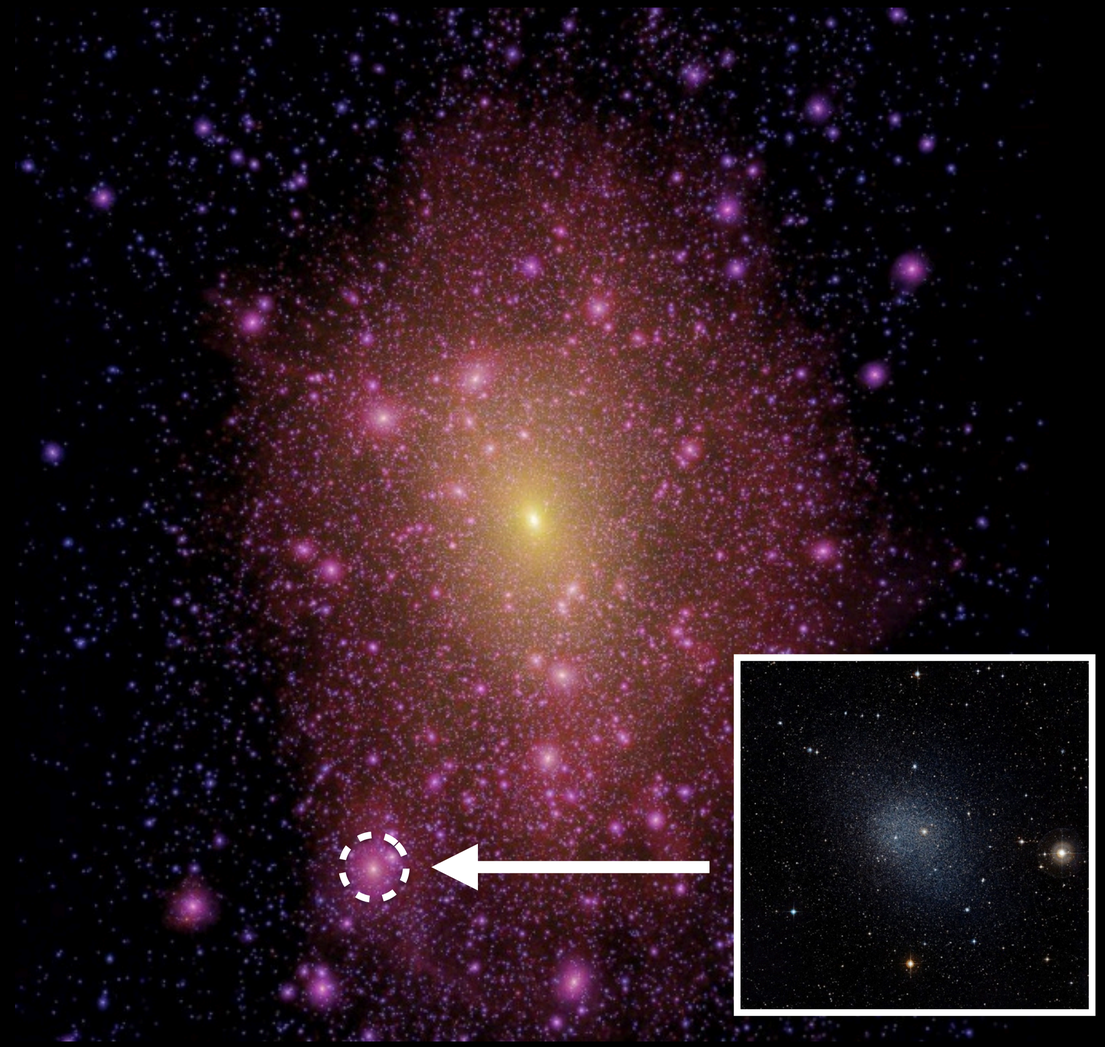
Galaxies Form in Dark Matter Halos



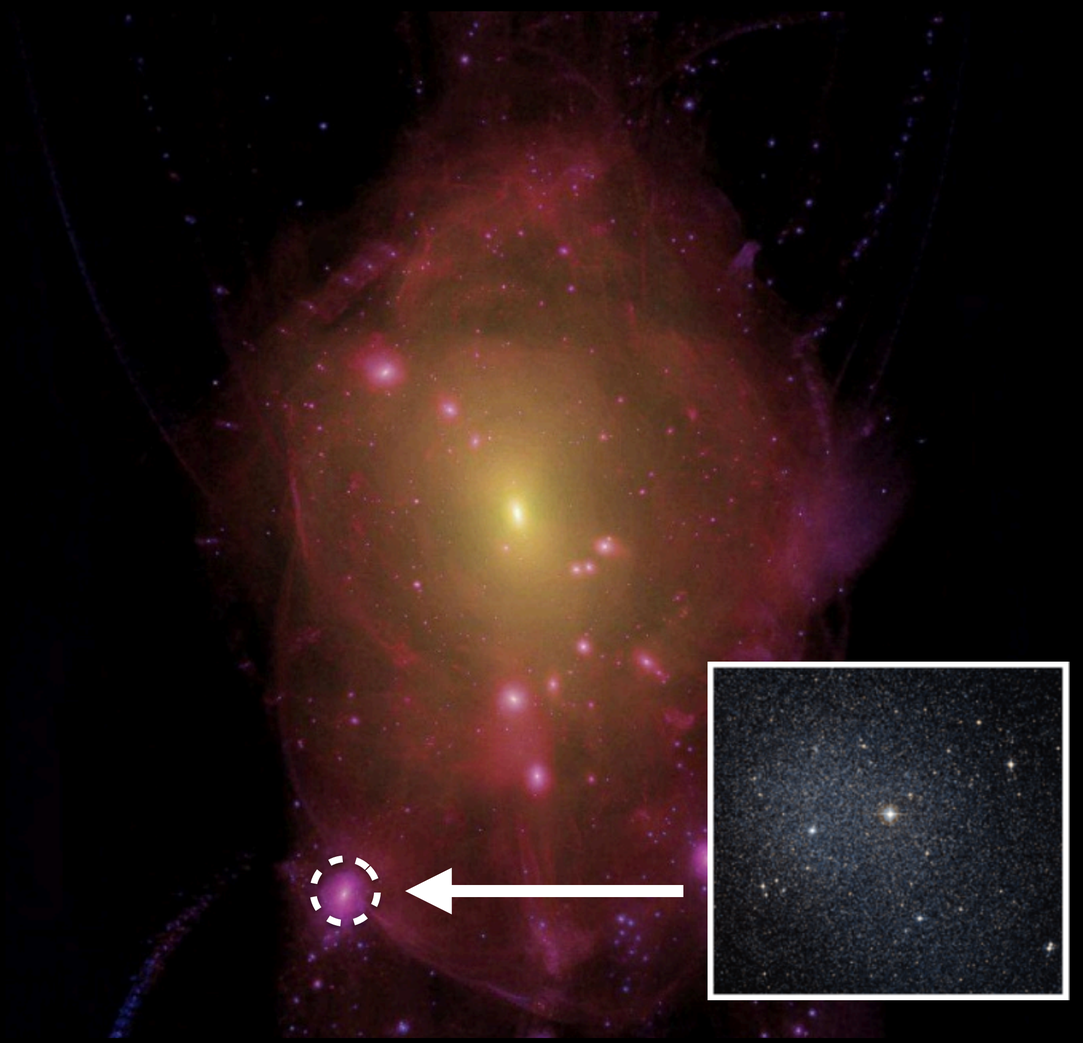


Dwarf Galaxies Test Dark Matter Models

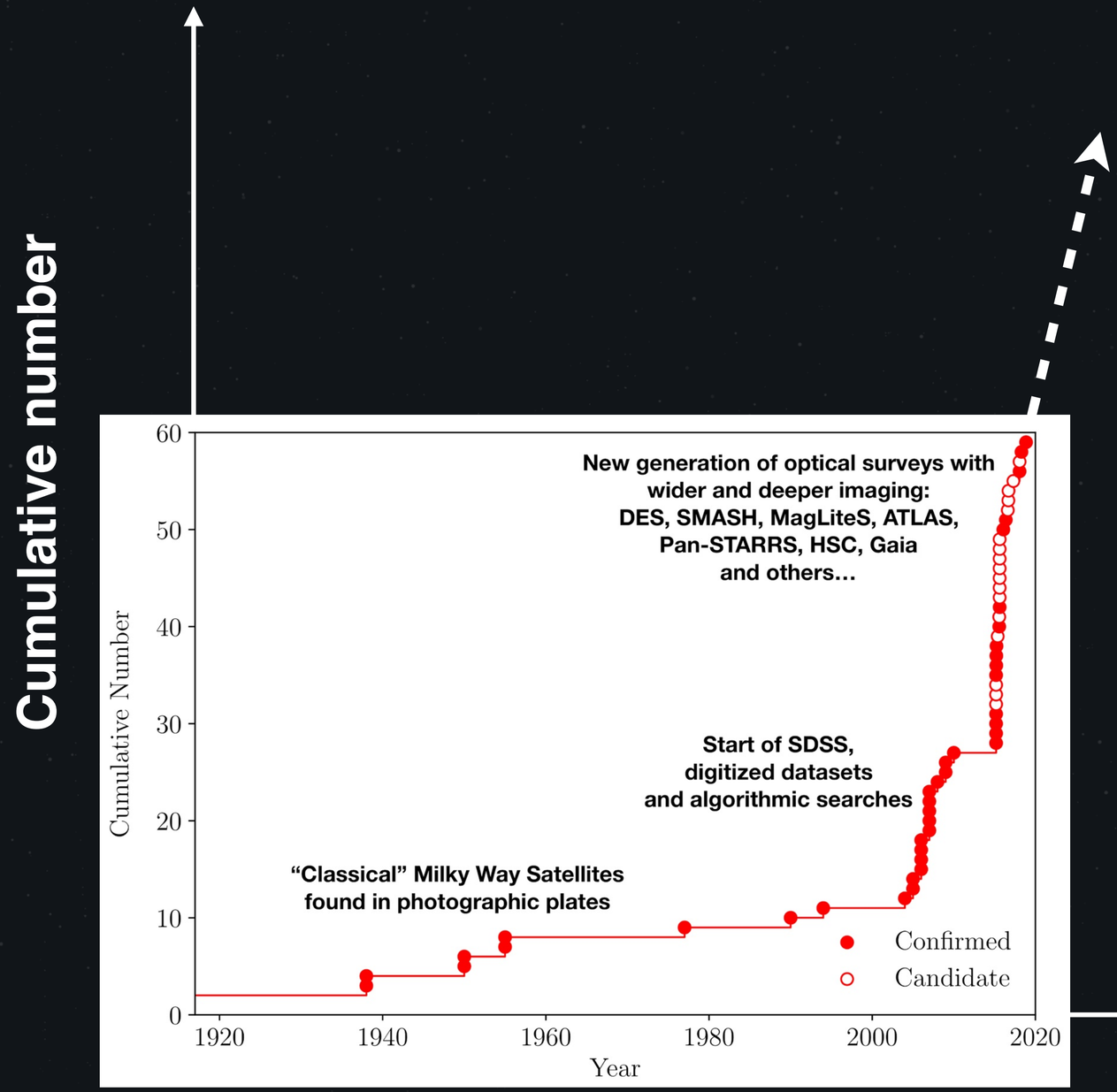
Cold Dark Matter

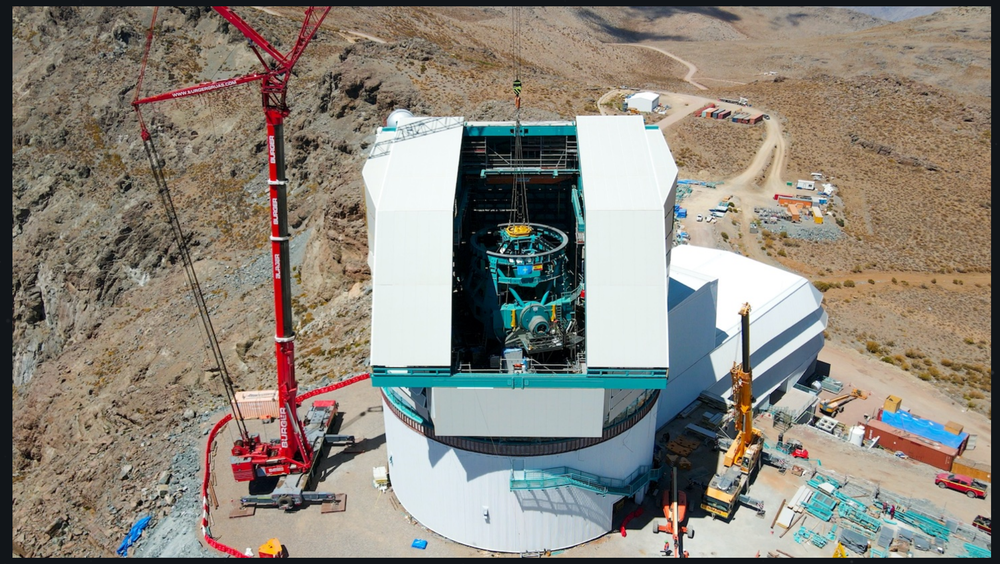


Warm Dark Matter



Upcoming Observations





Vera C. Rubin Observatory

Discussion

Discussion

Discuss the following topics with your neighbor:

- What do you find most interesting/surprising/confusing from today's material?
- What questions do you have about the universe based on today's material?